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SUPERSEDING
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MILITARY SPECIFICATION

SUPERSTRUCTURE, INTERIOR AND END BAY,

FOR MOBILE FLOATING ASSAULT BRIDGE/FERRY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers interior and end bay superstructures which assemble with superstructure transporters conforming to MIL-T-52703 to make up the Mobile Floating Assault Bridge/Ferry.

1.2 Classification. The superstructures shall be of the following types as specified (see 6.2):

Type I - Superstructure interior bay.
Type II - Superstructure end bay.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-S-781	- Strapping, Steel, Flat and Seals.
TT-P-1757	- Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity.
PPP-T-60	- Tape: Packaging, Waterproof.

FSC 5420

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Military

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-W-5044	- Walkway Compound, Nonslip, and Walkway Matting, Nonslip.
MIL-C-11796	- Corrosive Preventive Compound, Petrolatum, Hot Application.
MIL-P-12742	- Primer Coating: Phenolic, Water Immersible.
MIL-C-16173	- Corrosion Preventive Compound, Solvent Cutback, Cold-Application.
MIL-E-23765/2	- Electrode and Rod-Welding, Bare, Solid, Mild and Alloy Steel.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-M-52662	- Marine Drive Assembly and Marine Drive Installation Assembly for Bridge/Ferry Superstructure Transporter.
MIL-T-52703	- Transporter: Welded Hull, for Mobile Floating Assault Bridge/ Ferry.

STANDARDS

Federal

FED. STD. No. 595	- Colors.

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1246	- Product Cleanliness Level and Contamination Control Program.

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MS28759	- Hose Assembly, Rubber, Hydraulic, (3,000 PSI), Flared Tube.
MS39020	- Band, Marker, Blank.

DRAWINGS

ME

TA13207E6692	- Superstructure, Interior Bay: for Mobile Floating Assault Bridge/ Ferry.
TA13207E6689	- Superstructure, End Bay: for Mobile Floating Assault Bridge/Ferry.
D13207E6591	- Blocking of Interior Bay Superstructures (Railcar and Trailer Loading).
D13207E6592	- Blocking of End Bay Superstructures (Railcar and Trailer Loading).

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN WELDING SOCIETY, INC. (AWS)

A5.1 - Mild Steel Covered Arc-Welding Electrodes.
A5.5 - Low-Alloy Steel Covered Arc-Welding Electrodes.
A5.17 - Bare Mild Steel Electrodes and Fluxes for Submerged-Arc Welding.
A5.18 - Mild Steel Electrodes for Gas Metal-Arc Welding.
D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 North West Seventh Street, Miami, FL 33125.)

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Welding Qualifications, Section IX.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

STEEL STRUCTURES PAINTING COUNCIL (SSPC)

SP-6 - Surface Preparation Specification No. 6 Commercial Blast Cleaning.

(Application for copies should be addressed to the Steel Structures Painting Council, 4400 Fifth Avenue, Pittsburgh, PA 15213.)

3. REQUIREMENTS

3.1 Description. The superstructures and superstructure components shall be as shown on the applicable Top Assemblies (see 2.1), and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the supplier shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the superstructure. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative. The dimensions shown on the drawings apply at temperatures from 68° F to 70° F.

3.1.2 Gaged dimensions. Unless otherwise specified (see 6.2), all dimensions identified on the drawings by the symbol ⟨I⟩ shall be inspected using the Government-loaned inspection equipment (see 3.10), to determine acceptability in accordance with Quality Assurance Pamphlet P 715-106. The gages are calibrated at 68° F to 70° F.

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3.2 First articles.

3.2.1 Preproduction model. The supplier shall furnish one or more of each type of superstructure or separately procured superstructure component(s) as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his methods will produce superstructures or superstructure components that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examinations and tests shall be conducted by the supplier subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

3.2.2 Hydraulic-system components. In addition to the preproduction model, the following hydraulic components shall be furnished and tested by the supplier prior to the time the preproduction model is furnished: one each of all hydraulic fittings, hose assemblies, flow regulators, valves, quick disconnects, and hydraulic cylinders. The hydraulic components shall be identical to those furnished on the preproduction model. Tests shall be those specified in Section 4 and shall be subject to surveillance and approval by the Government.

3.3 Initial production. When specified (see 6.2), the supplier shall furnish to the Government one or more superstructures for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification. Aluminum materials shall be provided with the material manufacturer's name or trademark and the applicable alloy and temper designations in accordance with the respective ASTM specifications cited on the drawings.

3.4.1 Hydraulic oil. Oil for hydraulic systems shall conform to MIL-L-46167.

3.4.2 Sealant tape. Sealant tape for joints threaded with standard tapered pipe threads (NPT) shall be commercial type and shall withstand temperatures within the range of minus 65° F to plus 200° F. The tape shall be of the type which will allow the joints to be disconnected.

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3.4.3 Filter paper. Filter paper shall be 0.8 micron.

3.4.4 Nonskid coating. Nonskid coating shall conform to MIL-W-5044, Type I, Class II, Color No. 34087 in accordance with FED. STD. No. 595.

3.4.5 Welding electrodes.

3.4.5.1 For aluminum. Welding electrodes for aluminum shall be as specified on the applicable drawings.

3.4.5.2 For steel.

3.4.5.2.1 High-strength steel. Welding electrodes for high-strength steel shall conform to AWS A5.5, Classification E12015, E12016, E12018, E12018-M; or MIL-E-23765/2, Designation MIL-120S-1.

3.4.5.2.2 Low-alloy and structural steel. Welding electrodes for low-alloy and structural steel shall conform to the following AWS specifications and classes:

Specification	Electrode Classification	
AWS A5.1	E7015	7018
	E7016	7028
AWS A5.5	E7015A1	E7018A1
	E7016A1	
AWS A5.17	F72EM12 or F71EM12K	
AWS A5.18	Supplier's option	

3.5 Hydraulic system. The assembled hydraulic system of each type of superstructure shall withstand a working pressure of 3300 psi and a proof pressure of 5000 psi without external leakage (except weepage normally expected at piston rod packing), permanent deformation, or rupture of any part of the system. Joints threaded with standard tapered pipe threads (NPT) shall be sealed with commercial-type sealant tape. Prior to delivery, the hydraulic system shall be filled with hydraulic oil conforming to MIL-L-46167. The hydraulic functions of each superstructure shall start, operate, and stop in a smooth, controlled fashion, without external leakage, excessive binding, excessive chatter, erratic response, excessive vibration, or abnormal noise. The operating times for the hydraulic functions shall be within the time limits specified in Table IV.

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3.5.1 Component and system cleanliness.

3.5.1.1 Component cleanliness. Prior to installation onto the superstructures, all hydraulic components shall be washed and flushed throughout to remove all visible contaminants, particles, and foreign elements. Openings of the components shall be immediately covered to prevent the entry of contaminants (with the exception of the curb, pinning, rotation and unfolding cylinders). These cylinders shall be flushed until the amount of contaminant retained on 0.8 micron filter paper does not exceed 3 mg/100 ml of flushing fluid. All openings in the cylinders shall be immediately covered after reducing the components' cleanliness to the prescribed level. After installation, all components, except the cylinders, shall be flushed until the cleanliness level of this subsystem conforms to Table I.

3.5.1.2 System cleanliness. Cleanliness of the hydraulic system of all superstructures before and after completion of the tests specified in 4.6.2.13 and 4.6.2.16 shall conform to Table I.

Table I. Contamination Limits - Particles Per Milliliter

Quantity (max)	Particle Size
1600	Greater than 10 micron
114	" " 20 "
17	" " 30 "
3.8	" " 40 "

3.5.2 Hydraulic cylinders.

3.5.2.1 Rotation, pinning, and curb cylinders. The rotation, pinning, and curb cylinders for all types of superstructure(s) shall conform to the performance and dimensional requirements of the applicable drawings.

3.5.2.2 Unfolding cylinder. The unfolding cylinder for Type II superstructures shall be in accordance with the applicable drawings and shall conform to the following functional requirements:

(a) Operating pressure - 8000 psi, plus or minus 100 psi.
(b) Proof pressure - 12,000 psi minimum, maintained for 5 minutes.
(c) Cleanliness - The cylinder shall operate with hydraulic oil meeting the cleanliness requirements of MIL-STD-1246, Level 300.

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(d) Internal leakage - Leakage past the piston at 5 psi and 8000 psi shall not exceed 8 fluid ounces per minute.
(e) External leakage - No external leakage permitted (except weepage normally expected at piston rod packing).
(f) Endurance - Cycle the cylinder through 10,000 cycles of operation without failure.

3.5.3 Hydraulic fittings. Unless otherwise specified herein, all hydraulic components and fittings shall withstand a working pressure of 3000 psi and a proof pressure of 5000 psi without permanent deformation or rupture. The hydraulic fittings between the unfolding cylinder rod end ports and the check valves shall withstand a working pressure of 8000 psi and a proof pressure of 12,000 psi without permanent deformation or rupture.

3.5.4 Counterbalance and check valves. The counterbalance valves shall operate as specified up to a 3000 psi working pressure and shall be capable of being subjected to a proof pressure of 5000 psi. The pilot-operated check valves, except as specified herein for the pilot-operated check valves in the ramp fold/unfold circuit, shall operate as specified up to 3000 psi $\pm$ 100 psi, shall be capable of being subjected to a proof pressure of 5000 psi without evidence of permanent deformation or rupture, and have a flow of 10 gpm maximum. The pilot-operated check valve of the ramp fold/unfold circuit shall operate as specified up to a working pressure of 8000 psi and shall be capable of being subjected to a proof pressure of 12,000 psi. The pilot section of the counterbalance valve shall actuate under 25 percent of the load pressure, while the pilot section of the pilot-operated check valves, except as noted below for the pilot-operated check valves in the ramp fold/unfold circuit, shall hold at 600 psi (+ 25 psi, -0 psi). The pilot section of the pilot-operated check valves in the ramp fold/unfold circuit shall operate at 2500 psi (max). These valves shall also have a flow rate of 26 gpm at 40 psi drop and shall have an opening ratio of 5.0:1 and a decompression ratio of 36:1 with an unrestricted mounting position. All other check valves shall operate as specified up to a working pressure of 3000 psi and shall be capable of being subjected to a proof pressure of 5000 psi without any evidence of permanent deformation or rupture. All the valves shall function as specified with MIL-L-46167 as the operating fluid.

3.6 Lubrication. Prior to testing and delivery, the superstructures shall be lubricated as specified on Drawing 13207E6710.

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3.7 Cleaning, treatment and painting.

3.7.1 Cleaning.

3.7.1.1 Ferrous metal surfaces. Ferrous metal surfaces to be painted shall be blast cleaned in accordance with SSPC-SP-6 to remove all oil, grease, mill scale, products of corrosion, dirt, welding slag, or spatter, and other substances that may be detrimental to the applied coating system.

3.7.1.2 Aluminum surfaces. Aluminum surfaces to be painted shall be cleaned in accordance with MIL-T-704.

3.7.2 Treatment and painting. Each type of superstructure and each superstructure component(s) shall be treated and painted in accordance with MIL-T-704, Type B; finish coat shall be lusterless.

3.7.2.1 Dissimilar metals. Dissimilar metals in Type II superstructures, aluminum surfaces in contact with steel surfaces, shall be insulated from each other by a coat of zinc chromate primer conforming to TT-P-1757 or phenolic water-immersible primer conforming to MIL-P-12742.

3.7.3 Nonskid compound. After cleaning, treatment, and painting, nonskid compound shall be applied to the underside of the girders of the tapered end of the Type II superstructure.

3.8 Electrical wiring. Electrical wiring for the intercommunications system, when tested as specified in 4.6.2.14, shall exhibit no resistance exceeding 2 ohms for continuity or less than **1 megohm for ground.**

3.9 Identification marking for hose assemblies. Each hose assembly shall have identification marking metal-die stamped on an aluminum band conforming to MS39020 except that the band shall be of sufficient width to allow a minimum of three lines of 1/8-inch metal stamping. The band marking shall contain the following information:

MS28759-XXXXX
NSN 4720-XX-XXX-XXXX
Mfg. Code Date of Mfg.

3.9.1 Special identification marking for hose assemblies and superstructure fittings. Hose assemblies which connect the superstructure with the transporter hydraulic system shall be marked with the

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special identification marking shown on the drawings. The superstructure marking shall be metal-die stamped with 1/2-inch high numerals located adjacent to the respective fittings for hoses numbered 1 through 8 as shown on the drawings.

3.10 Government-loaned property. Unless otherwise specified (see 6.2), the following property, in the quantities indicated, will be loaned by the Government (see 6.4):

Item No.	Qty	Description	Identification
1	1 set (partial)	Inspection equipment for bridge/ ferry unit, interior and end bay.	Quality Assurance Pamphlet, P715-106.
2	1	Test frame: Superstructure, for mobile floating assault bridge/ ferry.	Drawings D93568-2195-1000.
3	1	QA Pamphlet "Care, Operation, and Maintenance of Inspection Equipment for Bridge/Ferry Unit, Interior Bay, and End Bay, Amphibious: Mobile Floating Assault Bridge/Ferry Unit".	P 715-106, included in Top Assembly microfilm.

3.11 Assembly and performance. Each assembled superstructure, when tested as specified in 4.6.2, shall perform as specified herein without malfunction, permanent deformation, or external leakage of hydraulic components.

3.12 Type I superstructure. Type I superstructures shall be as shown on Top Assembly TA13207E6692 and as specified herein.

3.13 Type II superstructure. Type II superstructures shall be as shown on Top Assembly TA13207E6689 and as specified herein.

3.14 Reliability and maintainability. The superstructure, when assembled on a transporter conforming to MIL-T-52703, shall (1) have a minimum **Mean-Time-Between-Failure** (MTBF) of 515 hours and (2) operate 500 hours without need of a major overhaul when operated with the transporter and a marine drive conforming to MIL-M-52662 in accordance with the established operational, maintenance, and safety procedures employed in MAB land, ferry, and bridge operations when tested as specified in 4.6.2.15.

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3.15 Workmanship.

3.15.1 Castings and forgings. All parts, components and assemblies of the superstructure which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce strength or affect function.

3.15.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges and corners are required or where they are not detrimental to safety.

3.15.3 Aluminum welding. The aluminum surfaces of parts to be welded shall be free from scale, paint, grease, oxide film, and other foreign matter. The welding shall be done by the inert gas (argon or helium) shielded-arc method. Welds may be ground, filed, buffed, or chipped but not hammered and shall be free from defects such as base metal cracks adjacent to or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, porosity, lack of penetration, lack of prescribed fit, inclusions, or spatter.

3.15.4 Steel welding. The surfaces of steel parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter other than mill scale that can be removed by chipping and wire brushing. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the superstructures to conform to the examination and test requirements specified in Section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch, unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the superstructures shall be at the option of the supplier.

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3.15.4.1 Care of welding electrodes. Care of welding electrodes specified for welding high-strength steel shall be as follows:

(a) When a container of welding electrodes is opened, the entire contents shall be placed immediately in a baking oven with electrodes placed not more than three layers deep on a tray and baked either (1) for a minimum of 1 hour, but not more than 1-1/4 hours at 800° F; or, (2) for a minimum of 4 hours at 400° F.
(b) The baked electrodes, while still warm, shall then be placed in a ventilated holding oven set at 250° F to 300° F.
(c) Not more than a 1/2-hour supply of welding electrodes shall be taken from the holding oven at one time.
(d) If electrodes are out of the holding oven in excess of 1/2 hour but less than 1 hour, they shall be returned to the holding oven for a period of not less than 1 hour.
(e) If electrodes are out of the holding oven in excess of 1 hour, they shall be returned to the baking oven and rebaked as described under (a).
(f) Electrodes removed for use shall not be allowed to come in contact with any moist or wet surfaces. Electrodes that have been in contact with water, grease, or oil shall not be used.

3.15.5 Welders. Before assigning any welder to manual welding of steel or aluminum welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1, Structural Welding Code, Section 5, Qualification.
Welding Qualifications of the ASME, Section IX.

Suppliers who make only horizontal welds need not qualify welders for "all position welding". The supplier shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS and ASME codes.

3.15.6 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. All fasteners shall be correctly torqued and shall have full thread engagement.

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3.15.7 Machine work. Tolerances and gages for metal fits shall conform to the limits specified on applicable drawings.

3.15.8 Jigs and fixtures. Components (except minor parts) shall be assembled in jigs or fixtures and joined while held in position. Jigs or fixtures shall be designed to minimize distortion of the parts being welded. Subsequent straightening and thermal stress relieving are permitted to relieve distortion, provided the straightening process does not damage the component.

3.15.9 Interchangeability (see 6.5.1). All like commercial components furnished by the supplier shall be directly interchangeable with respect to installation, performance and quality. The fabricated assemblies shall be interchangeable between superstructures.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.2 Disassembly inspection. Failure of any examination or test by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the preproduction model.

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4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of preparation for delivery (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction Type I and Type II superstructures or separately procured superstructure component(s) shall be examined for the defects specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests.

4.3.2.1 Hydraulic-system components. Prior to installation of the hydraulic-system components into the preproduction superstructure, these components shall be tested as specified in 4.6.2 (see Table III, Column 1). Failure of any test shall be cause for rejection or cause for performing the inspection specified in 4.1.2, as applicable.

4.3.2.2 Superstructure. After successful completion of the tests specified in 4.3.2.1, the hydraulic-system components shall be installed in the preproduction superstructure and the superstructure tested as specified in 4.6.2 (see Table III, Column 2). Failure of any test shall be cause for performing the inspection specified in 4.1.2.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production superstructures of each type as applicable, will be selected at random by the Government from superstructures being produced from production tooling and will be examined as specified in 4.6.1 and tested as specified in 4.6.2 (see Table III, Column 3) to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production superstructure shall not exclude the remaining superstructures from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any or all other tests contained in this specification as part of the initial production test, and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

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4.4.1 Inspection failure. Failure of an initial production superstructure to meet any requirements specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production superstructure(s) and shall be cause for refusal by the Government to continue acceptance of production superstructures until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on superstructures previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all superstructures accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for examination and tests for separately procured superstructure components shall be in accordance with MIL-STD-105.

4.5.2 Examination.

4.5.2.1 Individual. Each superstructure shall be examined for the defects specified in 4.6.1. Presence of one or more defects shall be cause for rejection of the superstructure.

4.5.2.2 Samples. Samples of separately procured components selected in accordance with 4.5.1 shall be examined for the applicable defects specified in 4.6.1. Acceptable quality level (AQL) shall be 2.5 percent defective.

4.5.3 Tests.

4.5.3.1 Individual. Each superstructure shall be tested as specified in 4.6.2. Failure of any test shall be cause for rejection.

4.5.3.2 Samples. Samples of separately procured components selected in accordance with 4.5.1 shall be subjected to the applicable test specified in 4.6.2. AQL shall be 2.5 percent defective.

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4.6 Inspection schedule.

4.6.1 Examination. Each superstructure shall be examined in accordance with Table II and as specified herein.

Table II. Examination Schedule

Preproduction			Initial production	Quality conformance	Defect		Requirement para.
Pretest	Posttest	Separately procured comp.					
1	2	3	4	5	6		7
X	-	X	-	X	101.	Nonconformance to any dimension marked (I) on the drawings.	3.1.2
X	-	X	-	X	102.	Nonconformance to any dimension or characteristic on the drawings not marked (I).	3.1.1
X	-	X	-	X	103.	Heat treatment not as specified.	3.1.1
X	-	X	-	X	104.	Commercial components not as specified.	3.1.1
X	-	X	X	X	105.	Age of hydraulic hose not as specified.	3.1.1
X	-	X	-	X	106.	Materials not as specified.	3.4
-	X	-	X	X	107.	External leakage, permanent deformation or rupture of any part of the superstructure hydraulic system.	3.5
X	-	-	-	X	108.	Nonconformance to cleanliness specified for individual hydraulic components.	3.5.1.1
X	-	X	-	X	109.	Rotation, pinning, and curb cylinder not in accordance with applicable drawing.	3.5.2.1

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Table II. Examination (Cont'd)

1	2	3	4	5	6	7
X	-	X	X	X	110. Superstructure lubrication fittings and lubrication not as specified.	3.6
X	-	X	X	X	111. Swage-locking pins and cadmium plating of swage-locking pins not as specified.	3.1.1
X	-	X	X	X	112. Cleaning, treatment and painting not as specified.	3.7
X	-	X	-	X	113. Insulation coat between dissimilar metals missing or not as specified.	3.7.2.1
X	-	-	X	X	114. Nonskid compound not applied properly or missing.	3.7.3
X	-	X	X	X	115. Hose identification not as specified.	3.9
X	-	X	X	X	116. Special hose identification not as specified.	3.9.1
X	-	X	X	X	117. Workmanship not as specified.	3.15
X	-	X	-	X	118. Qualification of welders not as specified.	3.15.5
X	-	X	X	X	119. Welding not as specified.	3.15.3, 3.15.4

4.6.1.1 Component inspection. Separately procured component tests may be performed at supplier's facilities provided all requirements of specifications are met.

4.6.1.2 Pretest examination. Pretest examination shall be done on individual components during fabrication, prior to assembly, or during assembly, where posttest examination shall be performed, as indicated, on those parts or requirements applicable to or affected by the test operations.

4.6.2 Tests.

4.6.2.1 Test conditions. Unless otherwise specified herein, all tests shall be conducted at an ambient air temperature between plus 40° F and plus 90° F, and hydraulic oil temperature shall be between plus 40° F and plus 190° F at the pump discharge. Hydraulic oil conforming to MIL-L-46167 shall be used for all tests.

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4.6.2.2 Test schedule. All tests shall be conducted in accordance with Table III and in the specified sequence.

Table III. Test Schedule

Preproduction: Hydraulic-system components	Preproduction: Superstructure	Initial prod.	Quality conf.	Test	Test para.
1	2	3	4	5	6
X	X	-	X	Test conditions.	4.6.2.1
-	X	X	X	Pretest flushing and cleanliness.	4.6.2.16
X	-	-	-	Limbering up of hydraulic cylinders.	4.6.2.3
X	-	-	-	Proof pressure.	4.6.2.4
X	-	-	-	Internal leakage.	4.6.2.5
X	-	-	-	Endurance.	4.6.2.6
X	-	-	-	Counterbalance valve.	4.6.2.7
X	-	-	-	Check valves.	4.6.2.8
X	-	-	-	Cleanliness of hydraulic components.	4.6.2.9
-	X	-	X	Rotation, curb, and pinning hydraulic circuits.	4.6.2.10
-	X	-	X	Ramp fold/unfold circuit.	4.6.2.11
				Superstructure operational test.	
-	X	X	X	Type I superstructure.	4.6.2.12
-	X	X	X	Type II superstructure.	4.6.2.12
-	X	-	X	Posttest cleanliness of hydraulic circuits.	4.6.2.13
-	X	-	X	Intercom system.	4.6.2.14
-	-	X	-	Reliability and maintainability.	4.6.2.15

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4.6.2.3 Limbering up of hydraulic cylinders. Prior to subjecting the cylinders to the tests specified herein, each cylinder shall be cycled through a minimum of 25 complete cycles. One cycle shall consist of extension and retraction of the rod through full travel without bottoming. The cycling shall be done by hydraulic pressure.

4.6.2.4 Proof pressure. Place each cylinder in a test frame so that the piston is restrained from bearing against either cylinder head. Connect the rod-end port and the cap-end port of the cylinder to a common hydraulic line so that both ends may be pressurized simultaneously. Apply the following hydraulic proof pressure to the cylinders:

(a) Curb, rotation and pinning cylinders - 5000 psi minimum to piston side.
(b) Unfolding cylinder - 12,000 psi minimum to piston side.

Evidence of leakage or permanent deformation shall constitute failure of this test.

4.6.2.5 Internal leakage. Connect each curb, rotation, pinning, and unfolding cylinder to a hydraulic system and place the cylinder in a horizontal position with the ports face up. The piston shall be unrestrained and unlimbered (see 4.6.2.3) to purge the system of air. Insert the cylinder in a test frame or fixture so that the piston is restrained from bearing against either cylinder head. Insert a graduated, transparent tube of either plastic or glass and fasten it vertically to the cap-end cylinder port in such a manner as to prevent leakage. One end of the tube shall be open to atmospheric pressure. Apply hydraulic pressure and maintain this pressure in the cylinder at the rod end. Allow the hydraulic oil bypassing the piston to enter and collect in the vertical tube. Measure the rise of fluid level during a period of not less than 3 minutes and determine the rate of internal leakage. Repeat this procedure on the cap end and measure the internal leakage. Subject the cylinders to the following pressures:

(a) Curb, pinning, and rotation cylinders at 3500 psi and 5 psi - measure leakage past the piston at both pressures on both sides of the piston.
(b) Unfolding cylinder at 8000 psi and 5 psi at the rod-end port - measure leakage past the piston on one side only.

Leakage in excess of the rate specified on the applicable drawings for the curb, pinning, and rotation cylinders, or in excess of the rate specified in 3.5.2.2 for the unfolding cylinder, shall constitute failure of this test.

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4.6.2.6 Endurance. Place each cylinder in a suitable test frame and cycle the cylinders through the following number of cycles by applying a hydraulic pressure of 3000 psi, plus or minus 100 psi, alternately to the cap- and rod-end port:

(a) Pinning, curb, and rotation cylinder - 25,000 cycles.
(b) Unfolding cylinder - 10,000 cycles.

The rate of cycling shall be 20 cycles per minute, plus or minus 5 cycles per minute, for the pinning, curb, and rotation cylinder and 2 cycles per minute for the unfolding cylinder. Any evidence of external leakage, except weepage normally expected at piston rod packing, binding of any part, permanent deformation, failure to cycle, or rupture of any part shall constitute failure of this test.

4.6.2.7 Counterbalance valve. Subject all sections of the counterbalance valves to a static proof pressure of 5000 psi. Maintain the pressure for not less than 5 minutes. Operate each counterbalance valve by establishing 1000 psi, 2000 psi, and 3000 psi against the check, then relieving this pressure by applying 22 to 28 percent of the checked pressure at the pilot port. Nonconformance to 3.5.4 shall constitute failure of this test.

4.6.2.8 Check valves. Subject all sections of the check valves, except the pilot-operated check valve of the ramp unfold circuit, to a proof pressure of 5000 psi. Maintain the pressure for not less than 5 minutes. Operate each check valve at a pressure of 1000, 2000 and 3000 psi. Check operation of the pilot-operated check valve by applying 22 to 28 percent of the checked pressure to the pilot port. Subject the pilot-operated check valve of the ramp unfold circuit to a test pressure of 8000 psi. Maintain this pressure for not less than 5 minutes. Subject the pilot port to 1600 to 2240 psi. Nonconformance to 3.5.4 shall constitute failure of this test.

4.6.2.9 Cleanliness of hydraulic components. Prior to assembling the hydraulic system, clean and flush each hydraulic component, including rigid lines and hoses. Prior to flushing, position each cylinder in such a manner that entrapment of contaminants is a minimum for each port. Filter the final flushing fluid for each component through 0.8 micron millepore filter paper. Reflush if necessary until the contaminant on the filter paper weighs no more than 3 mg/100 ml of flush fluid. Seal all openings in the component immediately after reducing the

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contamination to the acceptable level. Inability to conform to the cleanliness level specified in 3.5.1.2 shall constitute failure of this test.

4.6.2.10 Rotation, curb, and pinning circuits. Remove the clevis pins from the rod ends of the rotation and curb cylinders and position all cylinders in the rotation, curb, and pinning circuits so that their rods may move freely through their strokes. Connect the quick-disconnect couplings of each circuit to a supplier-furnished external hydraulic system capable of generating 5000 psi. Purge the system to remove all trapped air and cycle the cylinders of each circuit not less than 5 times to permit the entire circuit to be pressurized. Build up the pressure in each side of each circuit to 5000 psi and maintain this pressure for a minimum of 5 minutes. Nonconformance to 3.5 shall constitute failure of this test.

4.6.2.11 Ramp fold/unfold circuit. Remove the clevis pins from the rod ends of the unfolding cylinders and position the cylinders so that their rods may move freely through their entire strokes. Connect the quick-disconnect couplings to a supplier-furnished external hydraulic system capable of generating 5000 psi. Purge the system to remove all trapped air and cycle the cylinders not less than five times to permit the entire circuit to be pressurized. Build the pressure in each side of the circuit to 5000 psi and maintain this pressure for a minimum of 5 minutes. Nonconformance to 3.5 shall constitute failure of this test.

4.6.2.12 Superstructure operational test. Unlimber each type of superstructure by placing the superstructure on the Government-furnished test frame and subject the superstructure to 25 complete operational cycles which shall include the following:

Interior Bay (Type I)

Bridge Half Cycle

(a) Lock holddown bolts
(b) Unlock holddown bolts
(c) Raise superstructure.
(d) Rotate superstructure.
(e) Lower superstructure.
(f) Lock holddown bolts.
(g) Raise curbs.
(h) Extend forward and aft locking pins.

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Stow Half Cycle

(a) Lower curbs.
(b) Retract locking pins.
(c) Unlock holddown bolts.
(d) Raise superstructure.
(e) Rotate superstructure.
(f) Lower superstructure.

The bridge half cycle and stow half cycle constitute one full cycle. Repeat this bridge half cycle and stow half cycle in sequence for a total of 25 complete operational cycles (locking and unlocking of holddown bolts required on only one complete operational cycle).

End Bay (Type II)

Bridge Half Cycle

(a) Lock holddown bolts.
(b) Unlock holddown bolts.
(c) Raise superstructure.
(d) Rotate superstructure.
(e) Lower superstructure.
(f) Lock holddown bolts.
(g) Extend aft locking pins.

Stow Half Cycle

(a) Unlock aft locking pins.
(b) Unlock holddown bolts.
(c) Raise superstructure.
(d) Rotate superstructure.
(e) Lower superstructure.

The order of operation shall be reversed to complete one complete cycle.

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Fold/Unfold Operations
(End bay in bridge position)

(a) Attach test stand counterweight tie rods to end bay and lock holddown bolts.
(b) Unfold Ramp.
(c) Raise curbs.
(d) Lower curbs.
(e) Fold ramp.
(f) Unlock holddown bolts.
(g) Detach counterweight tie rods when 25 cycles are completed.

The bridge half cycle and stow half cycle constitute one full cycle. Repeat the bridge half cycle and stow half cycle in sequence for a total of 25 complete operational cycles (locking and unlocking of holddown bolts required on only one complete operational cycle). After the bridge and stow cycles have been completed, perform 25 cycles of the fold/unfold operation (b) - (e).

Table IV. Operating Time for Hydraulic Functions

Function	Elapsed Time (Secs)
Stow-to-Bridge	25
Bridge-to-Stow	28
Curbs Up (Int)	14
(End)	16
Curbs Down (Int)	20
(End)	16
Pins, Extend	6
Pins, Retract	6
End Bay, Unfold	40/65
End Bay, Fold	40/75

NOTE: The above elapsed operating times can be demonstrated on a MAB transporter (MIL-T-52703) at 1500-1550 engine rpm, or on the superstructure test frame equipped with a hydraulic pump delivering 25 to 32 gpm at 500 psi.

4.6.2.13 Posttest cleanliness of hydraulic circuits. After completion of the superstructure operational tests and prior to delivery of the superstructures to the Government, remove samples of hydraulic

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fluid from the quick-disconnect coupling of each leg of each circuit. Test these samples for the particle count specified in Table I (see 3.5.1.2). If the particle count exceeds the values in Table I, disconnect the hydraulic lines at the cylinders and continue to flush and sample until the contamination is reduced to the acceptable level. Inability to conform to the cleanliness levels specified in Table I shall constitute failure of this test.

4.6.2.14 Intercom system. The intercom harness shall be checked for grounding of any wire to each other or to the shield. Any ground having a resistance of less than **1 megohm shall constitute failure** of this test. The harness shall be checked for continuity for each wire and shield. Continuity shall be exhibited throughout the structure including the transporter leads. Any resistance found exceeding 2 ohms shall constitute failure of this test.

4.6.2.15 Reliability and maintainability. These tests will be performed by the Government at site(s) selected by the Government. The tests will be conducted with the superstructure(s) mounted on transporter(s) equipped with the specified on-vehicle materiel (OVM), consistent with the established operational, maintenance, and safety procedures employed in the MAB land ferry and bridge operations as specified in 4.6.2.15.1 and 4.6.2.15.2. The MTBF for the superstructure shall be calculated by dividing total MAB system operating time (see 4.6.2.15.1 and 4.6.2.15.2) by the number of superstructure reliability failures. A superstructure reliability failure shall be an occurrence which:

(a) Precludes or prevents the continuation of specified water operations;
(b) May result in permanent damage or deformation to the system if not corrected;
(c) Reduces the performance characteristics of the system to a level below specified minimum requirements; or
(d) Presents a safety hazard to operating personnel through continued operation.

Any failure occurring during prescribed operations which is correctable within 20 minutes by MAB crew members using on-board tools and materials shall not be utilized in MTBF calculations. A failure shall be charged if an incipient or previously undetected malfunction is discovered during preventive maintenance and this malfunction would have previously been considered chargeable if the need had arisen to operate the affected

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system prior to scheduled maintenance. An incipient malfunction detected during the correction of another malfunction shall also be charged provided the incipient failure is unrelated to the first malfunction, requires maintenance to prevent progression to the point of malfunction, and it satisfies the scoring criteria listed in (a) through (d). Incidents resulting from test item abuse, unrealistic operating conditions, or accidents unless such are a direct result of malfunctions of the test item shall not be chargeable failures.

4.6.2.15.1 Operational land. The operational land tests shall be conducted in accordance with Table V.

Table V. Operational Land Tests

Course	Mileage and Speeds
Paved roads	1,000 miles, varying speeds up to maximum.
Gravel and dirt roads	1,000 miles, varying speeds up to maximum.
Level cross country	1,000 miles, varying speeds up to maximum safe speed.
Hilly cross country	1,000 miles, varying speeds up to maximum safe speed.

4.6.2.15.2 Operational water. Operate the superstructure assembled on an equipped transporter in bridge, ferry and hydraulic slaving operations for a minimum cumulative total of 200 hours. The testing of each unit shall include operations within the following specified limits:

(a) At least 950 and not more than 1050 simulated ferry crossings (85% of the crossings shall be with a Class 60 vehicle).
(b) At least 1330 and not more than 1470 vehicle crossings in the bridging mode (25% of the crossings shall be with a Class 50 vehicle or greater).
(c) At least 22 and not more than 40 fold/unfold cycles of the end bay superstructure.
(d) At least 22 and not more than 40 bridge hookups.
(e) At least 22 and not more than 40 superstructure bridge/stow cycles.

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4.6.2.16 Pretest flushing and cleanliness. Prior to any operations, actuations, or testing, the superstructure hydraulic system with the cylinders hydraulically disconnected and not included in the flushing circuit, shall be flushed and tested for cleanliness. After flushing, remove samples from any one quick-disconnect coupling in each of the four hydraulic circuits and analyze for conformance to Table I. If the particle count exceeds the values in Table I, continue flushing and sampling, with the cylinders hydraulically disconnected, until the contamination is reduced to the acceptable level. After the specified cleanliness is reached in all circuits, connect the cylinders to the hydraulic system and continue testing. Nonconformance to the requirements of Table I shall constitute failure of this test.

4.7 Inspection comparison. The Government may select superstructures at any time during the contract production period and subject these superstructures to the examination specified in 4.6.1 and the tests specified in 4.4 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production superstructures. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison superstructure to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison superstructures and shall be cause for refusal by the Government to continue acceptance of production superstructures until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on superstructures previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all superstructures accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

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4.8 Inspection of preparation for delivery.

4.8.1 Quality conformance inspection.

4.8.1.1 Unit of product. For the purpose of inspection, a completely processed Type I or Type II superstructure shall be considered a unit of product.

4.8.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.1.3 Examination. Samples selected in accordance with 4.8.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

120.	Materials not as specified. Each incorrect material shall constitute one defect.
121.	Curbs not placed in the retracted position as specified.
122.	Exposed portions of piston rods not coated with preservative as specified.
123.	Coated surfaces of piston rods not covered with barrier material as specified, and the wraps not secured with tape as specified.
124.	Unconnected fittings not covered with tape as specified.
125.	Unconnected lines not secured to the superstructure as specified.
126.	Machined ferrous metal surfaces not coated with preservative as specified.
127.	Coated machined surfaces not covered with barrier material as specified, and the wraps not secured with tape as specified.
128.	Intercom connecting fitting not covered with tape as specified.
129.	Intercom cable and preserved publications not secured to the superstructure as specified.
130.	Type II superstructure not folded and secured as specified.
131.	Steel straps not applied as specified.
132.	Strapping not Class B or Level A.
133.	Marking illegible, incomplete, incorrect or missing for Level A or C.

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5. PREPARATION FOR DELIVERY

5.1 Preservation. Preservation shall be Level A (see 6.6).

5.1.1 Hydraulic system. Prior to preservation, curbs shall be placed in the retracted position.

5.1.1.1 Piston rods. Exposed portions of piston rods shall be coated with preservative conforming to MIL-C-11796, Class 3, applied in accordance with MIL-P-116, and wrapped with barrier material conforming to MIL-B-121, Type I, Grade A, Class 1 or 2. The wrap shall extend approximately 2 inches onto the piston rod cylinder and shall be secured in place with tape conforming to PPP-T-60, Type IV.

5.1.1.2 Fittings and lines. Unconnected fittings shall be covered with waterproof tape specified in 5.1.1.1. Unconnected lines shall be secured to the superstructure with soft wire. The wire shall be placed on a fitting rather than on the line itself.

5.1.2 Miscellaneous preservation. Unpainted, unmachined ferrous metal surfaces shall be coated with preservative conforming to MIL-C-16173, Grade 1, applied in accordance with MIL-P-116. Machined ferrous metal surfaces shall be coated with preservative as specified in 5.1.1.1 and then wrapped or covered with greaseproof barrier material which shall be secured with waterproof tape as specified in 5.1.1.1. The intercom connecting fitting shall be covered with waterproof tape and the intercom cable shall be secured to the superstructure with soft wire above the girder bottom flange level.

5.1.3 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, Method IC-1. The preserved publications shall be secured to the superstructure for which they are intended in a conspicuous but protected location.

5.2 Packing. Packing shall be Level A or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Type I. The Type I superstructure shall be shipped uncrated as a unit. Curbs shall be secured with two steel straps as shown on Drawing D13207E6591.

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5.2.1.2 Type II. The Type II superstructure shall be folded and secured in that position with the slinging link assembly. Two steel straps completely encircling the superstructure shall be used to prevent the curbs from extending as shown on Drawing D13207E6592. The Type II superstructure shall be shipped uncrated as a unit.

5.2.2 Level C.

5.2.2.1 Type I. The Type I superstructure shall be packed as specified for Level A.

5.2.2.2 Type II. The Type II superstructure shall be packed as specified for Level A except that steel strapping shall be Type I, Class A, conforming to QQ-S-781.

5.3 Loading. The Type I superstructures shall be loaded for shipment in accordance with Drawing D13207E6591. The Type II superstructures shall be loaded in accordance with Drawing D13207E6592.

5.4 Marking. Marking for shipment shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Type I and Type II superstructures are intended to be assembled with transporters conforming to MIL-T-52703. The superstructures and transporters when assembled form a basic interior bay or an end bay of the Mobile Floating Assault Bridge/Ferry. The transporter furnishes transportation for the superstructure on land and in the water and is also the floating support for the bridge or ferry. The completed floating bridge is designed to carry a Class 60 track or wheel vehicle load or combination for an expected life cycle of approximately 15,000 vehicle crossings.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of superstructure required (see 1.2).
(c) When inspection of dimensions identified by the symbol (I) other than as specified is required (see 3.1.2).
(d) Time frame required for submission of the preproduction model and hydraulic components and number of superstructures (see 3.2.1 and 3.2.2).

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(e) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the supplier (see 3.2.1).
(f) When initial production inspection is required and number and types of superstructures required (see 3.3).
(g) When Government-loaned property is not required (see 3.10).
(h) Welding process required when other than as specified (see 3.15.4).
(i) Level of packing required (see 5.2).

6.3 Preproduction model. Any changes or deviations of production superstructures or separately procured superstructure component(s) from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish superstructures conforming to this specification.

6.4 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.10.

6.5 Definition.

6.5.1 Interchangeability. Interchangeability is defined as two or more components possessing such functional and physical characteristics as to be equivalent in performance and durability and capable of being exchanged one for the other without alteration of the components themselves or of adjoining components except for adjustment, and without selection for fit or performance.

6.6 Preservation of superstructure. Preservation specified in 5.1 is the minimum required to protect the superstructures against hazards of worldwide distribution and indefinite periods of outdoor storage. It is also considered that the preservation specified is the minimum which will assure protection for the superstructures when they are exposed to the least severe conditions of shipping and storage. Preservation which is applicable to more than one level of shipping and storage conditions is normally specified only for the most severe conditions for which it is adequate. That method of preservation should also be used for all less severe conditions.

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6.7 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts list, and supplier's maintenance and operation manual to be furnished with each superstructure.

6.8 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools) and instructions on shipment of the superstructures.

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Preparing activity:

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)CUMENT IDENTIFIER AND TITLE MIL-S-52355D(ME) Superstructure, Interior and End Bay, For Mobile Floating Assault Bridge/Ferry

\ME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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